SafePack

Safe & Sustainable Packaging Made Simple

Choose the right packaging based on your food properties. Get scientific recommendations in under 3 minutes.

SCIENTIFIC — GREEN — COMPLIANT

"Helping MSMEs bridge the gap between food science and packaging technology."

TELL US WHO YOU ARE

Manufacturer
Optimizing production lines

Farmer / FPO
Value addition & shelf life

Professional
Research & Compliance

Start Analysis →

Home | Projects | Standards | Profile

New Analysis

FOOD PROPS | CONDITIONS | SAFETY | REVIEW

Step 1: Food Properties

"Science-driven inputs for accurate estimation."

Water Activity (a_w)

Dry (0.2) **0.85** Moist (1.0)

Degradation Sensitivities

- [x] Oxidation (Oxygen Sensitive)
  Lipid or Vitamin C loss
- [ ] Moisture Loss/Gain
  Crispness or hydration
- [ ] Light Sensitivity
  Photo-oxidation & pigment loss

Anticipated Shelf-Life Goal

3 Months | **6 Months** | 12 Months

Primary Market Climate

Tropical (Hot & Humid)

Save Draft

Next: Storage Conditions

Analysis Results

AI TOP MATCH

Bio-Based PLA Barrier

SHELF LIFE
185 Days

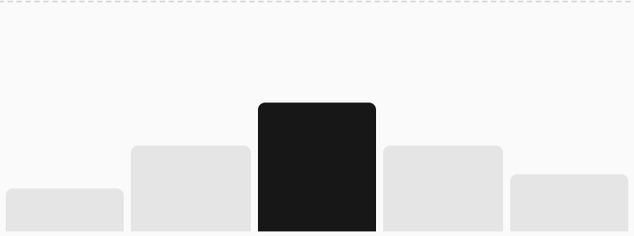
SUSTAINABILITY
A+

"What-If" Analysis

LIVE SIMULATOR

Storage Temp (Avg) **28°C**

Barrier Thickness **45 µm**



PREDICTED SHELF-LIFE VARIANCE

Material Comparison

Bio-PLA (Eco)
Compostable • High O2 Barrier
180d
$0.12/pc

Recycled PET
Recyclable • Standard Barrier
120d
$0.08/pc

Adjust Inputs

Finalize Choice

COMPLIANCE | PASSPORT

Safety Gate Check

"Model Estimates are separate from Safety Checks."

Migration Testing (Overall)
Limits conform to EU 10/2011 standards.

Specific Migration (SML)
Checked for BPA, Phthalates, and NIAS.

Heavy Metal Content — UPLOAD DOC
Lead, Mercury, Cadmium & Chromium limits.

PACKAGING PASSPORT

ID: SP-2026-X8

Material	**Bio-PLA Blend**
Carbon Footprint	**0.45 kg CO2e / kg**
End-of-Life	**Industrial Compost**

"Transparency builds trust with your eco-conscious consumers."

Export Technical Dossier

Email Certificate to Compliance Team

Sustainable End-of-Life

"What happens when the food is gone?"

UPCYCLING & WASTE HIERARCHY

1. Human Consumption (Surplus)

If product exceeds shelf-life but is safe: Donate to food banks or secondary markets.

FIND LOCAL FOOD BANKS

2. Animal Feed

Processing side-streams for livestock or poultry feed conversion.

3. Industrial Composting

Bio-based packaging and food scrap co-management. Returns nutrients to soil.

CONSUMER DISPOSAL GUIDE

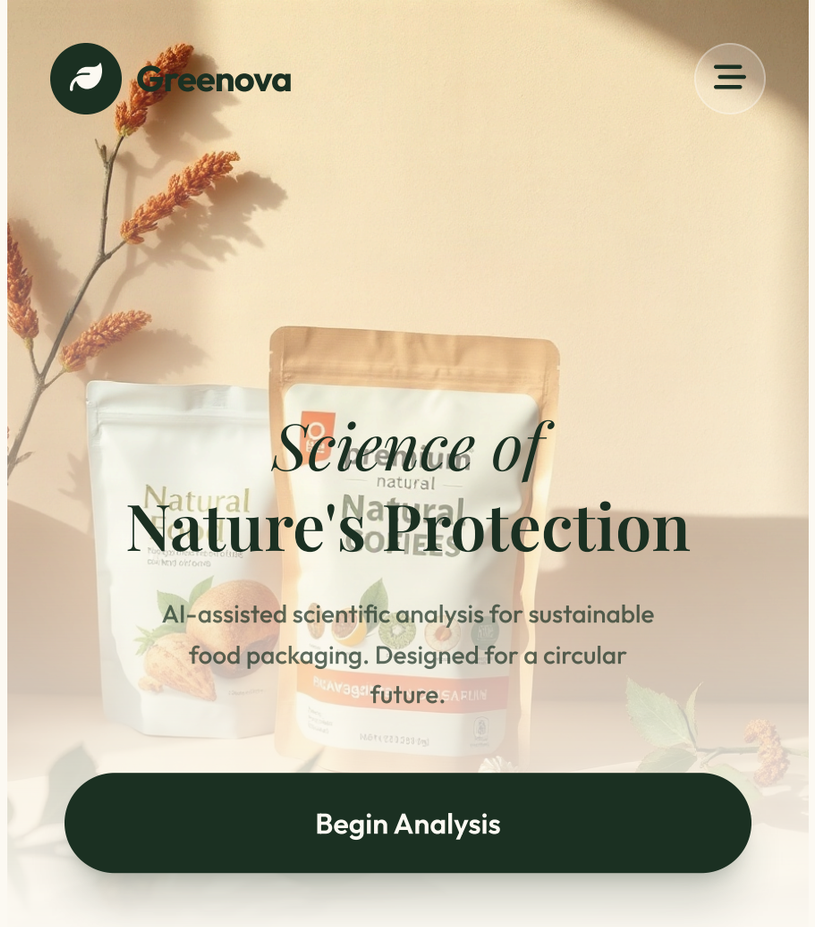
REMOVE LINER

RINSE PACK

Note: Upcycling recommendations are based on standard food safety guidelines. Always verify with local regulatory bodies for specific food-grade byproduct repurposing.

Generate Disposal Instructions

Greenova



Science of Nature's Protection

AI-assisted scientific analysis for sustainable food packaging. Designed for a circular future.

Begin Analysis

Food Categories

VIEW ALL









Precision Compliance

Our model separates AI-assisted ranking from regulatory safety gates, ensuring every choice is both green and safe.

EXPLORE THE SCIENCE

Defining Food Properties

Precise scientific inputs are the foundation of accurate packaging recommendations.

CHEMICAL ENVIRONMENT

Water Activity (a_w) **0.82**

DRY (0.2) MOIST (1.0)

Acidity (pH Level) **4.5**

ACIDIC NEUTRAL ALKALINE

SENSITIVITY ANALYSIS

Oxygen (Oxidation)
Lipid oxidation or Vitamin C degradation risk.

UV & Visible Light
Photosensitive pigment or vitamin loss.

Ethylene Gas
Accelerated ripening in climacteric produce.

VISUAL MATERIAL REFERENCE



TOP SCIENTIFIC MATCH

98.2 COMPATIBILITY SCORE

Bio-PLA Fiber Coated Mesh

SHELF LIFE
180 Days

IMPACT
A++

"Optimized for climacteric respiration and ethylene control."

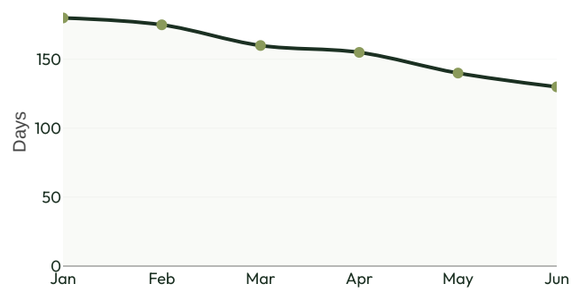
DETAILS

"What-If" Analysis

INTERACTIVE SIMULATION TOOL

Storage Temperature 22°C

Oxygen Barrier (OTR) 15.5



ALTERNATIVE OPTIONS

02 **Recycled Kraft Laminate** Score: 84.5
120d High Eco VIEW

03 **Compostable Barrier Film** Score: 79.2
95d Bio-Based VIEW

Proceed to Compliance

Regulatory Verified

SAFETY GATE STATUS: PASSED

VERIFICATION CHECKLIST

Global Migration Testing
Conforms to EU 10/2011 and FDA standards for food contact materials.

Ref: ISO 9001 · Doc #2921-A

Specific Migration Limits (SML)
Safety assessment for heavy metals, BPA, and phthalates complete.

Local Regulatory Variance
Pending review for Southeast Asian regional specific additives.

ASK REGULATORY Q&A

PACKAGING PASSPORT

Digital Certificate

UUID: GN-8821-X-2026

SUSTAINABILITY RANK	Excellent (A++)
CARBON IMPACT	0.42 kg CO_2e / kg
MATERIAL BASE	Lignin-Based Polymer
DISPOSAL ROUTE	Home Compostable

"This digital passport provides end-to-end transparency for supply chain partners and consumers."

DOSSIER

Next: Circular Economy



A Life Beyond Use

SUSTAINABILITY STRATEGY

RECOMMENDED PATHWAYS

SECONDARY LIFE
Priority: High

Repurpose agricultural side-streams for non-food grade industrial fiber production or sustainable construction binders.

EXPLORE PARTNERS

NUTRIENT CYCLE
Priority: Medium

Convert bio-residuals into nutrient-rich soil amendments through managed industrial composting systems.

STANDARD PROTOCOL

CONSUMER DISPOSAL GUIDE

- REMOVE ADHESIVES
- RINSE & DRY PACK
- CHECK LOCAL SORTING
- HOME COMPOST

Finalize Strategy